

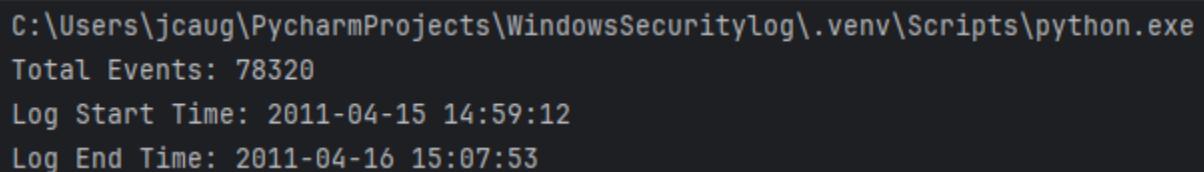
BLUF

This is an analysis of a Windows Security Log that contains approximately 80,000 events over 24 hours, 155 distinct users, and two indicators of compromise. Two main indicators of compromise were abnormal user logon patterns and suspicious failed logon attempts. While these are not inherently malicious, their presence poses a risk that warrants further investigation. Analysis of this security log revealed logons during all hours of the day for users Grant Larson and Matt Edwards. This could indicate compromised credentials or automated activity, which could be a breach of security and policy. Two failed login attempts from the 127.0.0.1 IP address (Event ID 4625) will require additional information. Users who used the machine at 127.0.0.1 around 15:00 UTC on April 15th, 2011, must be evaluated to determine whether the traffic was generated by legitimate administrators or bad actors.

An overview of the data you were given.

I obtained an overview of the data using a Python script. This script was “vibe-coded” by ChatGPT when I asked it to write a program to parse the data and provide metrics on it.

Fig. 1 - Python Script Output

A screenshot of a terminal window with a dark background and light-colored text. The text shows the execution of a Python script to analyze a Windows Security Log. The output includes the file path, the total number of events, and the start and end times of the log data.

```
C:\Users\jcaug\PycharmProjects\WindowsSecuritylog\.venv\Scripts\python.exe
Total Events: 78320
Log Start Time: 2011-04-15 14:59:12
Log End Time: 2011-04-16 15:07:53
```

As shown in Figure 1, there are 78,320 events in the Windows Security Log file. The start time of the log information is April 15th, 2011, at 14:59 (or 10:59 AM eastern time), and the end time

is April 16th, 2011, at 15:07 (or 11:07 AM eastern time). The total capture time of this security log is 24 hours and 8 minutes.

Fig. 2 - Python Script Output (User Summary)

```
=== 4624 User Summary ===  
Distinct Users: 155
```

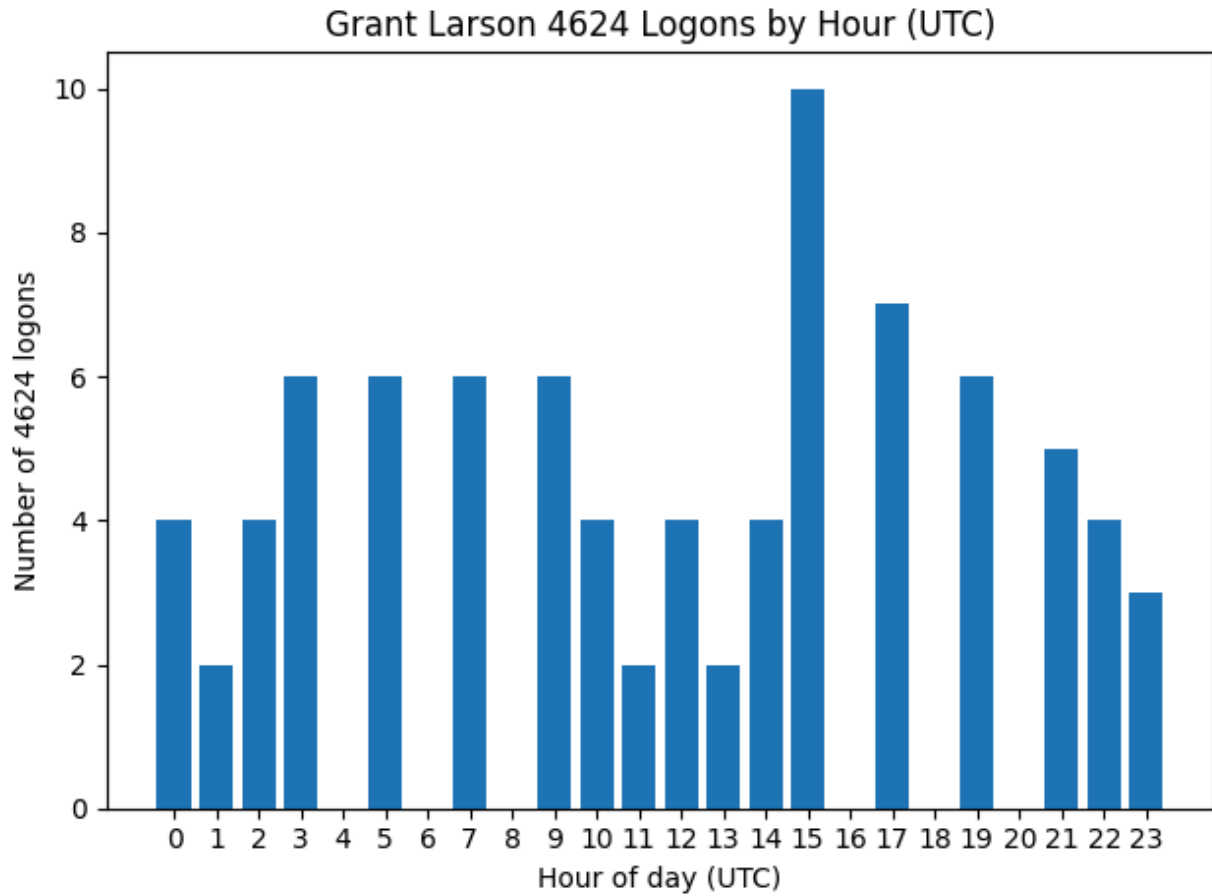
As shown in Figure 2, there are 155 distinct users in the Windows Security Log.

Fig. 3 - Python Script Output (Top Users)

```
AFC\grant.larson: 85  
AFC\Matt.Edwards: 85  
AFC\randal.graves: 84  
AFC\shauna.leedy: 83  
AFC\hannibal.smith: 83  
AFC\don.allen: 83
```

As seen in Figure 3, Grant Larson and Matt Edwards are tied for the most logons.

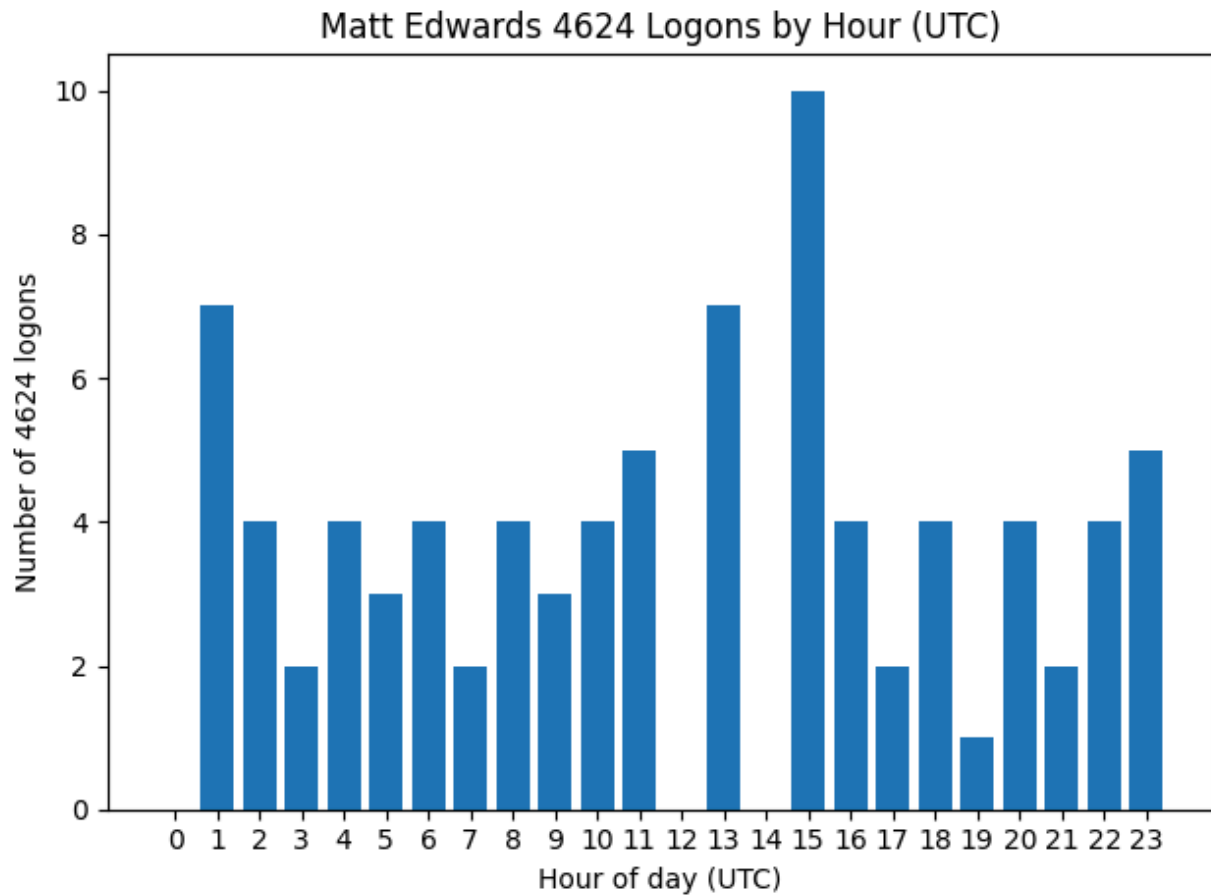
Fig. 4 - Frequency Chart of Grant Larson Logons.



As shown in Figure 4, Grant Larson logs in at all times of day (most often at 11:00 AM EST).

This may be a potential indicator of compromise, but it is hard to tell since we may require additional information from the organization.

Fig. 5 - Frequency Chart of Matt Edwards



As seen in Figure 5, Matt Edwards is also logging in at all times of the day (most often at 11:00 AM EST). This could also be an indicator of compromise; however, it is difficult to tell without additional information from the organization that originated this security file.

Fig. 6 - EventID 4625 Total

```
Total 4625 Failed Logons: 2
```

As seen in Figure 6, there are 2 events with ID 4625. This event ID indicates a failed logon attempt and provides the TargetUserName, DomainName, IPAddress, and WorkstationName.

Fig. 7 - 1st event with ID 4625.

```
Time (UTC): 2011-04-15 15:05:01.875000+00:00
Computer: DC01.AFC.com
TargetUserName: administrator
TargetDomainName: AFC.com
LogonType: 2
IpAddress: 127.0.0.1
WorkstationName: DC01
```

Figure 7 provides many details about the first occurrence of a 4625 Event. The login attempt to access the “administrator” username was made on the AFC.com domain from the workstation DC01, which had an IP address of 127.0.0.1, and was executed within the [AFC.com domain](#).

Fig. 8 - 2nd event with ID 4625

```
Time (UTC): 2011-04-15 15:04:57.671875+00:00
Computer: DC01.AFC.com
TargetUserName: administrator
TargetDomainName: AFC.com
LogonType: 2
IpAddress: 127.0.0.1
WorkstationName: DC01
```

Figure 8 shows the second occurrence of a failed logon attempt event. The only detail that differs from the 1st occurrence is the time— this second occurrence was only 4 seconds earlier. This is a clear indicator of compromise, which must be investigated further. To solve this issue, we must implement a login timeout mechanism. This protocol would restrict or prohibit further logins for TargetUserName after several attempts. Additionally, multi-factor authentication would provide an extra safety net of security if a bad actor guesses the administrator's password.

Fig. 9 - PyCharm Python Script

```
XML_FILE = "SecurityLog-rev2.xml"
OUTPUT_FILE = "output.xml"

try:
    with open(XML_FILE, "r", encoding="utf-8") as source:
        content = source.read()

    with open(OUTPUT_FILE, "w", encoding="utf-8") as destination:
        destination.write(content)

    print(f"Successfully copied {XML_FILE} to {OUTPUT_FILE}")

except FileNotFoundError:
    print("Error: Source file not found.")
except Exception as e:
    print(f"An error occurred: {e}")
```

Figure 9 is a simple script that will copy and paste the target XML file “SecurityLog-rev2.xml” into a new file named “output.xml”.

Fig. 10 - Revised PyCharm Script

```

XML_FILE = "SecurityLog-rev2.xml"
OUTPUT_FILE = "output.xml"

try:
    with open(XML_FILE, "r", encoding="utf-8") as source:
        content = source.read()

    events = content.split("<Event ")
    filtered = [e for e in events if "<EventID>4624</EventID>" in e]

    with open(OUTPUT_FILE, "w", encoding="utf-8") as destination:
        for e in filtered:
            destination.write("<Event " + e if not e.startswith("<xmlns") else e)

    print(f"Successfully wrote {len(filtered)} EventID 4624 entries to {OUTPUT_FILE}")

except FileNotFoundError:
    print("Error: Source file not found.")
except Exception as e:
    print(f"An error occurred: {e}")

```

Figure 10 is the same Python script, but edited to copy and paste only events with ID 4624.

Fig. 11 - Revised PyCharm Script

```

Successfully wrote 34415 EventID 4624 entries to output.xml

Process finished with exit code 0

```

Figure 11 shows the output of the same Python script, with added functionality to count events with ID 4624.

Fig. 12 - Revised PyCharm Script

```
2011-04-15T14:00 - 37
2011-04-15T15:00 - 3546
2011-04-15T16:00 - 3299
2011-04-15T17:00 - 3685
2011-04-15T18:00 - 2999
2011-04-15T19:00 - 3264
2011-04-15T20:00 - 2939
2011-04-15T21:00 - 3101
2011-04-15T22:00 - 3008
2011-04-15T23:00 - 3225
2011-04-16T00:00 - 3301
2011-04-16T01:00 - 3521
2011-04-16T02:00 - 3268
2011-04-16T03:00 - 3413
2011-04-16T04:00 - 3118
2011-04-16T05:00 - 3426
2011-04-16T06:00 - 2917
2011-04-16T07:00 - 3512
2011-04-16T08:00 - 3090
2011-04-16T09:00 - 3388
2011-04-16T10:00 - 2893
2011-04-16T11:00 - 3327
2011-04-16T12:00 - 3019
2011-04-16T13:00 - 3225
2011-04-16T14:00 - 3290
2011-04-16T15:00 - 509
```

Figure 12 is the output of the same Python script, with the addition of displaying how many times events with ID 4624 occurred within an hour. This script will start at the beginning of the security log and go to the end. For this specific output, the file starts on April 15th, 2011, 14:00 UTC and ends on April 16th, 2011, 15:00 UTC.

To conduct this analysis, I mainly used PyCharm, a Python IDE, and ChatGPT. Within a PyCharm project, it is very easy to create a Python script and house an XML file within the same

folder. This proved helpful, as I could easily tweak the script without having to redefine the file's location. ChatGPT increased the overall efficiency of my workflow by allowing me to give it a prompt specifying what to change in a Python script, and it would return the desired output.

Indicators of Compromise (IOC's)

There are two main indicators of compromise within this Windows Security Log. The first being the login times and frequencies of users Matt Edwards and Grant Larson. These two users were logging in frequently outside of business hours, as clearly illustrated in Figure 4 and 5. While these login times alone are not inherently malicious, the two users should be investigated further.

The second indicator of compromise is the failed login attempts, as illustrated in Figures 7 and 8. The concern stems from the TargetUserName in both of the Figures, which is "administrator". Additionally, the two events occurred 4 seconds apart, clearly indicating that a user was attempting to guess the administrator's password but failed. Individuals who were using the machine at 127.0.0.1 around 15:04 UTC on April 15th, 2011, should be investigated immediately.